Dohun Kim

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Education

Weston High School — Weston, MA
High School Diploma (in progress) — GPA 4.0 UW — *2024 – Present*

Coursework & Exams: AP Physics C: Mechanics and E&M (A, 5); AP Calculus BC (5); AP Computer Science A (5); AP Statistics (A, 5); AP World History (A, 5).

Hanyang University — Gifted & Talented Center for Informatics — Seoul, South Korea
Advanced Course in Informatics — *2020 – 2021*

- Graduated #1 of 95 and youngest in the cohort; sole student selected for a 1-on-1 AI course under Prof. Ki Hyuk Sung.

Research & Publications

MIT PRIMES STEP — Senior Group, Mathematics Research — *2024 – 2025*

Co-authored two papers through MIT's selective research program, with original results in combinatorics and discrete structures.

- *Mathematics of Gozinta Boxes* — arXiv:2508.18277
- *Chip-Firing in Infinite k-ary Trees* — arXiv:2501.06675

Leadership & Activities

FIRST Tech Challenge — 22489 Galactic Narwhal Chicken Effect (Diamond) — *2024 – 2026*

- Co-Captain (2025–2026) and team member (2024–2025); led strategy, build, and software, and designed the drivetrain and clutch power-take-off lift in open-source CAD (Onshape).
- Team earned **2nd place for the Motivate Award** at the World Championship (Jemison Division).
- **Multinational Tech Invitational 2026:** won the **inaugural Mechatronics Synergy Award**.
- **Michiana Premier Event 2026:** Croft Division Finalist (alliance partner) and **1st place for the Control Award**.

Volunteer FLL Mentor — Teams "Butterfly Effect" & "LegoImpossible" — *2024 – 2026*
Weston Robotics — Planning Council & Program-Wide Mentor — *2026 – Present*
Weston Town Criers (A Cappella), Concert Choir & Jazz Choir; Tennis — *2024 – Present*

- Featured soloist on "500 Miles" (2025–2026); selected for MMEA Senior Districts as bass (2024 & 2025).

Honors & Awards

- **AMC 10A:** 141/150 (2024) • **AIME II:** 9/15 (2025) • MAML Olympiad Level 1: 2nd in Massachusetts (24/25).
- **USACO:** advanced to Gold Division, 900/1000 (2026) • **Codeforces:** rating 1737.
- IMLEM: #1 individual ranking with perfect scores across all 5 meets (2022–2023); MITIT 2023: 2nd, Beginner Division (team).
- Korean Olympiad in Informatics: Silver Award (National 2021; Regional 2021 & 2022).

Selected Projects

- **DSIM** (playdsim.com) — **Founder & Main Programmer** of an online 2D FTC driving simulator with solo, co-op and ranked 1v1/2v2 play: authoritative WebSocket server, deterministic physics/scoring sim, and a Glicko-2 ranked ladder. ~39k lines of TypeScript/React; 527 passing tests.
- **Empower Initiative** (empowerinit.org) — Website Coordinator for a student-run tutoring nonprofit; built and maintain the organization's public site.
- **A11y Checker** (demo) — web accessibility checker; **2nd place**, 2021 Congressional App Challenge (Office of Rep. Katherine Clark).
- **Voya** (voya.dohunkim.xyz) — interactive choose-your-own-adventure study game, built with Next.js.

Skills

Programming Languages: C/C++, JavaScript/TypeScript, Python, Java
Languages: English (Fluent), Korean (Fluent), Spanish (Intermediate)
Frameworks & Tools: React, Next.js, Onshape (CAD)